Literature Review

Ella Thorburn
University of Cape Town
Department of Computer Science
South Africa
threll002@myuct.ac.za

Abstract

Mobile health interventions for maternal and child health have been deployed across a growing number of lower and middle income countries, but the literature consistently identifies a lack of community involvement, cultural adaptation, and attention to the infrastructural realities of end users in their design. This review examines mHealth for maternal health in low-resource settings, focusing on the challenges and opportunities relevant to developing a culturally adapted mobile application for Sepedi-speaking women in rural Limpopo, South Africa. We will analyze findings from scoping reviews, co-design studies, information source research, and nutrition literature to identify core requirements for an effective intervention. The review identifies challenges we will face during development and testing, such as the conflict between needing to involve fathers in maternal health and the suppression of women's voices in workshops, and the absence of published dietary research for Sepedi-speaking communities. These findings will inform the design of our application for our Honours project: a storage conscious, offline-first mobile application that uses digital storytelling to deliver culturally grounded health content to pregnant women and their support structures in rural Limpopo.

Keywords

Maternal Health, mHealth, User-Centred Design, Low-Resource Settings, GRAVID-Adapt, Community Healthcare Workers (CHWs)

1 Introduction

Despite recent improvements in child and infant mortality across Sub-Saharan Africa, the area still lags behind developed countries, highlighting the continuing need to find solutions for this issue to address this issue[1]. Malnutrition in pregnant woman in lower-middle income countries (LMICs) remains a cause of concern, as tension often arises between accepted cultural practices and researched-backed western dietary recommendations [15] [4]. Lack of support and understanding from male partners and family members also remains a key issue for expecting mothers[12]. Mental health in expecting mothers has also been raised as a concern [12].

In South Africa, mobile health (mHealth) interventions have shown promise in alleviating some some of these issues.[14] The development of MomConnect, a message-based system developed in South Africa, aimed to deliver maternal health content and encourage antenatal care (ANC) visits. It achieved remarkable uptake and has shown promise in improving rates of clinic visits and being a useful source of information for mothers [3] [14]. Pregnancy +, a UK developed mobile app for pregnancy tracking and information access, is also mentioned in the literature, but has been identified as presenting significant challenges to mothers in South Africa, as it requires large amounts of storage to install on one's smartphone and requires internet access for updates and resource access [12].

Existing mHealth interventions show a lack of community involvement, cultural adaption, and little emphasis on nutritional information [17][1]. A need has been identified in South Africa for a mHealth solution that serves the needs and fits into the cultural context of its end users.[5]

For our Honours project, we aim to implement the first iteration of the solution that has been identified: A lightweight, offline-first mobile application aimed at improving the health and well-being of pregnant women in low-resource, low-income contexts. Our application will specifically serve Sepedi speaking women and their support structures in a rural Limpopo. The app will utilize digital storytelling to promote actionable learning of the delivered health content and utilize community collaboration to produce content that end-users feel they can trust. It will also provide term-specific reminders for appointments and nutritional advice. The app will take care to include other members of a mother's support structure and be tailored to their cultural context and language. The development of the application will include User Acceptance Testing of a mature MVP in the early stages and leverage adapted user co-design to iterate effectively on the feedback.

2 Body of Review

2.1 mHealth for Maternal and Child Well-being in LMICs

Mobile health interventions for maternal and child health have been deployed, with mixed success, across a growing number of lower and middle income countries (LMICs). India (20%), Ghana (9%), and China (9%) are leading the rollout of digital health interventions, which most commonly use mobile applications to deliver their content [1]. Adusei-Mensah examined 63 studies and found that many interventions failed to consider the cultural, social and infrastructural realities of their targeted end users [1].

A scoping review of 141 articles by Till et al. revealed similar findings, identifying that community involvement in the design of Maternal and Child Health (MCH) interventions in LMICs has been largely negligible [17]. The review found that at least one MCH project failed specifically because expert-led design did not account for community realities [17]. The review also found the inclusion of maternal mental health content and the engagement of families to be pertinent to the success of mHealth interventions [17] . The authors identified custom-developed mobile applications as a promising alternative to SMS-based systems (like MomConnect), noting their potential to deliver tailor-made interventions using images, video, text, and voice to increase the effectiveness of mHealth content delivery[17]. Further benefits to mobile applications were

identified such as reducing costs compared to USSD and SMS by the using cached or locally stored content in the application[17].

South Africa's very own MomConnect is an interesting example of an mHealth intervention that has achieved widespread usage, despite having many of the flaws identified by the scoping reviews [14]. MomConnect enrolled over 494,000 pregnant women in its launch year in 2014 and became one of only five maternal mHealth programs globally to scale beyond one million users [3]. By 2017, MomConnect reached over 60% of pregnant women at their first antenatal appointment and operated across 95% of public health facilities nationally [3]. Reflections by Peter and Barron identified that the WhatsApp and USSD messaging system that MomConnect utilizes has limited its ease of use and proved costly as it has scaled [3] [14]. An urgent need was identified to develop a custom mobile application to reduce costs, allow flexibility of features, and allow cost-effective multimedia content delivery [14].

The state of mHealth for MCH in rural Limpopo is well documented by Muthelo [12], whose findings directly led to our project and will inform our design choices. The research was conducted in the tribal community halls of three rural villages outside Polokwane and included mothers, Community Health Workers (CHWs), and community leaders, who were all fathers [12]. In the workshops participants highlighted that existing digital tools like MomConnect and Pregnancy+ were helpful for receiving updates on fetal development and potential complications, but were fell short in meeting their needs due to language barriers and resource constraints[12]. Some mothers expressed that they still preferred consulting traditional healers because of their cultural beliefs and practices [12]. The tension between digital health tools and traditional care is a recurring theme in the literature and one that our project will have to navigate carefully. Participants identified significant barriers to using existing apps such as data costs related to MomConnect and storage requirements for Pregnancy+ . One mother specifically reporting that she could not download the application because her phone did not have enough space [12]. Poor network connectivity and data affordability across the rural area further contributed to the issues in accessing digital health content from these sources [12]. When the participants were prompted to identify what they believed would be an effective solution, they expressed a desire for a single integrated app that covered both antenatal and postnatal care and had video calling to reach healthcare professionals [12]. Participants showed a strong preference for video and audio content over text and noted that people are more drawn to watching videos than reading, and that recorded voice and video could spread information more effectively [12]. They also wanted the app to be straightforward and usable even after the baby was delivered, without requiring additional downloads [12]. Importantly, the study also highlighted the role of fathers and traditional healers. The authors recommended that fathers should be involved and trained in maternal and child health, as their participation could support healthier pregnancies and child development [12]. They also suggested that incorporating traditional healers into the design of digital health apps could improve adoption and sustainability in rural areas [12]. These findings will directly inform our design choices, pushing us towards features such as offline-first content delivery, audio and visual educational material, storage-conscious app size, and the inclusion of family members and community figures in the application's design.

2.2 Co-design for mHealth in LMICs

The scoping reviews highlight the need for the involvement of end users in the development of mHealth applications, and the research conducted Coleman [5] actioned on this need. The study conducted generative co-design workshops with parents and community leaders across rural and urban sites, one of which was in rural Limpopo, and identified ten key MCH challenges, including parent well-being, tensions between traditional healers and western medicine, misinformation, accessible health information, and involving fathers [5]. Across all research sites, the top three priorities selected by participants were parent's emotional well-being, building parenting skills in early life, and accessible and affordable health information [5]. In the rural Limpopo sites emotional well-being was ranked as the highest priority, followed by parenting skills [5]. This finding aligns with the emphasis on maternal mental health in the scoping review by Till [17], and reinforces the need to include mental health resources in our application. A particularly encouraging finding for our project was that communities are open to and would welcome digital MCH interventions, as long as they are developed with their cultural and infrastructural contexts in mind [5]. Desire was also expressed for the intervention to be adapted to their home language [5]. The authors stressed that designing for African languages cannot rely on automated translation services like Google Translate, as it fails to convey intended meaning in most South African languages. They advocate for having first-language speakers involved in text and content creation from the start, as word-for-word translations often fall short when trying to convey meaning and may leave the content feeling unfamiliar [5]. This presents a significant challenge for our project, as we will have to find a willing collaborator to assist us with the language in our application and its content. Coleman also noted the importance of involving local intermediaries in community research [5], which identifies the need for a trusted community member to be present during our UAT.

An important question raised by the literature is not just what information mothers receive, but how they learn from it. Farao [7] argues that most digital health interventions have prioritised getting information to mothers over helping them actually understand and act on it. Farao proposes digital storytelling as a way to bridge this gap, suggesting that using digital technologies to tell stories grounded in existing cultural practices could make health information more relatable and actionable for mothers in under-resourced communities [7]. Although some audio and visual material has been provided to us by our proposer, this paper outlines a promising approach for developing more effective content for our project. We will have to work around the challenge identified that mothers are hesitant to share health information or personal experiences for fear of gossip and stigmatisation within their communities. [7]. Farao's research also further reinforces the findings of the literature, identifying that MCH encompasses more than just physical health, and that family conflict, feelings of exclusion, and stigmatization all contribute to how mothers engage with health information they receive[7]. This further motivates for the development of a maternal

health application that covers factors related to pregnancy above and beyond just physical well-being.

2.3 Designing, Adapting and Testing mHealth Applications

The mHealth intervention literature consistently identifies that prioritzing user-centered design and community involvement produces digital solutions that are better adopted and more usable than those that do not [17][1].

Farao [8] identified the need for a framework suitable for the design and adaption of mHealth applications in low-resource settings and proposed a framework that combines Information Systems Research (ISR) with design thinking. Farao identified that the ISR framework and design thinking fall short when applied to the developing world context, and proposed a novel approach which emphasizes empathy and leverages the strengths of each framework to address the shortcomings in the other. The study used the Post-Study System Usability Questionnaire to assess user satisfaction with the application after completing a task. This adapted framework proved to be successful when adapting a mHealth application used by CHWs and advocates that it would be effective when used in design too [8]. The framework falls short in delivering guidelines to develop application content for end users and does not emphasize co-design as much as the literature suggests is necessary for the development of a successful maternal and child well-being intervention.

A study by Molefe and Dlamini [11] provided clear evidence in a South African mHealth context that incorporating user-centered design and developing for the end-user's reality significantly improves adoption rates (72.4% compared to 41.2%) and 90-day retention (51.8% compared to 29.7%) when compared to the baseline application [11]. Molefe and Dlamini used iterative co-design and Human-Centered Design (HCD) to guide their development stage and utilized the System Usability Scale to assess their improved application. The researchers identified that low-data, offline-first, privacy-by-design interventions are key to adoption and retention of end users in a South African mHealth context [11].

Atakunda's [2] research in rural Uganda provided further proof on the effectiveness of iterative co-design in the development of mHealth interventions. The study aimed to develop a patient-centered mHealth intervention to encourage and support women to use maternity care services, and leveraged existing research to inform the design process [2]. The researchers conducted interviews with 30 women and 5 healthcare providers to identify their key antenatal care (ANC) topics and preferred content delivery format, then developed SMS text and audio content with the help of healthcare professionals. These findings were then used to develop and conduct user acceptance testing on a pilot application, culminating in a mature application that incorporated the user feedback. Throughout the research the women highlighted a need for information they knew they could trust amidst their limited and divergent information sources [2]. They believed that this information could help debunk misinformation myths over time [2]. The women also expressed that the messages sent by the application helped initiate discussions with their partners regarding their pregnancy [2]. At the end of the study, the participants put forward what they believed to be five key elements for a successful application: (1) education and preparation, (2) motivation, encouragement and support, (3) customisation and connection, (4) easy-to-use interface, and (5) engagement and empowerment. Although these findings may not be from our project's direct context, the study identified overlapping challenges between their participants and our end users. This observations suggests that these five points may serve as useful guidelines for the development of our application. At the end of the study, researchers highlighted an opportunity to include features in the application which encourage self-goal attainment and affirmations could likely motivate long-term use, an opportunity also highlighted in a study by Nkwo [13].

The study by Nkwo [13] surveyed 600 women across Nigeria in order to try understand what prevented or discouraged women from seeking maternal healthcare. Religious and cultural beliefs, as well as conflicting information from family members were identified as key issues[13]. The author suggested the use of Persuasive Systems Design as a method to improve MHC mHealth applications, by adding features that support and motivate users to perform target behaviours, such as engaging with maternal health educational content and attending antenatal appointments [13].

While the papers discussed above advocate strongly for community involvement in the design process, actually doing this work in culturally diverse settings can be challenging. Till and Farao [16] reflected on co-design workshops they conducted across four South African provinces and found language barriers and culturally embedded hierarchies to be direct challenges to workshops. In Limpopo specifically, the presence of community leaders and members of the local royal family brought complicated power dynamics into the workshop[16]. Female participants and younger people we often observed looking to senior figures during brainstorming and prototyping activities, as opposed to freely expressing their opinions and contributing to the prototypes [16]. The authors suggest that when discussing sensitive health topics it may be necessary to separate men and women, as they observed women in the study seemed restricted to speak freely in front of male participants[16]. This poses a direct conflict with the need that Muthelo [12] highlighted to include fathers in the development, and will pose as a difficult challenge to navigate when we perform our UAT . While we will not be implementing a co-design workshop ourselves, our UAT will involve community members from a similar Limpopo context, and the lessons from the reflection from Till and Farao regarding language, power dynamics, and how sessions are structured will shape how we conduct testing to make sure we are hearing honest and useful feedback from everyone in the room.

2.4 Information Overload: Voices Mothers Hear

Pregnant women in low-resource settings are exposed to health information from a myriad of different sources [6][12]. Healthcare professionals, family elders, traditional healers, other mothers, and increasingly, digital tools all deliver often conflicting information to these women [6][12] [17]. Understanding who expecting mothers listen to, why they trust them, and how they decide which advice to follow is essential for delivering an application with health content that will lead to positive actions for their well-being.

Dsane-Nsor [6] conducted interviews and co-design workshops that involved pregnant women, family members, and healthcare professionals in Techiman, Ghana and found that women rely on a diverse range of stakeholders, both formal and informal. The women emphasized a desire for information that could help them gauge whether their experiences were considered "normal" and provided guidance during the complexities of pregnancy [6]. Older women such as mothers, grandmothers, and mothers-in-law were identified as key sources for antenatal health information, with the participants expressing that they perceived them as a trustworthy source due to their personal experiences [6]. Reliance on this informal information source was seen to create tension, as the advice of these trusted sources often contradicts what healthcare professionals recommend [6]. This tension between formal and informal information sources was also identified by Muthelo [12] during research in rural Limpopo. Amongst these voices, Dsane-Nsor highlighted the value and potential of the internet and pregnancy apps for delivering learning tools and educational content [6]. The authors also however identified the need for these digital interventions in rural communities to include culturally relevant dietary practices rather than generic advice, as this would likely lead to better reception and less conflict with community information sources [6]. Dsane-Nsor also argued that mHealth interventions in this context should prioritise leveraging existing community support structures rather than attempting to replace them [6]. Findings from the study indicate that incorporating testimonials and narrative storytelling could be an extremely effective way to strengthen trust in digital information and improve reception of the intervention [6]. This finding connects directly back to Farao's [7] argument for using digital storytelling to help mothers move from just receiving health information to actively learning from and acting upon it.

Félix et al. [9] also examined parental experiences with multiple information sources, placing emphasis on figuring out how parents assess and filter the advice they receive. Through 14 workshops with 42 parents across Portugal and South Africa, the researchers found that parents do not just passively accept health information from any given source. Instead, they were found to actively evaluate advice based on several factors, such as whether it aligns with their own values and beliefs, whether it is feasible within their daily reality, and whether the person giving the advice is perceived as competent and well intentioned [9]. The study found that advice that felt generic, judgmental, or disconnected from their daily realities was more likely to be dismissed, even when it came from healthcare professionals [9]. Parents were observed to place the greatest trust in information from people who knew their child's history, who had gone through similar experiences, and who demonstrated empathy and a genuine willingness to help [9]. The researchers used these finding to inform four recommendations for mHealth interventions to increase trust and value placed on its information. These were: tailoring content to the child's characteristics, adapting content to medical history, acknowledging parents' values and beliefs rather than overriding them, and complementing text based messages with multimedia content [9]. The limitation of this study was that it addressed message based mHealth systems and did not directly involve pregnant women. However the recommendation for multimedia content to be utilized and for parents values and beliefs to be acknowledged to improve information uptake by parents is encouraging, as it further motivates the culturally adapted, multimedia approach of our application.

Both of these papers highlight that information delivered by unrelatable or impersonal sources is likely to be ineffective, regardless of its accuracy. Information tailored to the users context, and delivered through videos and audio from trusted voices have been identified as a key way to improve the effectiveness of mHealth content. These insights will be central to how our application approaches its content design and delivery.

2.5 The Nutrition Factor

Cultural beliefs and food taboos play a significant role in shaping what pregnant women eat across sub-Saharan Africa, often restricting access to foods that are nutritionally important during pregnancy. Chakona[4] studied 224 isiXhosa women in the Eastern Cape of South Africa and found that 37% followed food practices shaped by their cultural beliefs. The foods most commonly avoided included meat, fish, eggs, potatoes, fruits, and beans, all of which are rich in essential nutrients needed during pregnancy [4]. Most women had learned these taboos from their own mothers or grandmothers, but social groups we also mentioned as a source of disinformation [4]. Ramulondi [15] documented similar patterns when studying women in Northern Kwa-Zulu Natal, finding that traditional food taboos during pregnancy are deeply embedded in cultural beliefs and are passed down generations through families and communities. These practices are notably not simply matters of ignorance but are deeply tied to beliefs about protecting the health of the mother and the unborn child, even when the restricted foods are have been proven by health care professionals to be nutritionally beneficial [15].

What makes these findings particularly challenging for our project is that food taboos are culturally specific, and that findings from one cultural group cannot be assumed to apply to another. While research has documented these practices among isiXhosa women in the Eastern Cape [4] and in Zulu women in Kwa-Zulu Natal [15], no equivalent published study exists for Sepedi speaking communities in Limpopo Province. There is a risk we may have to rely on informal information sources to develop the nutrition section, as without any locally grounded understanding, our application's nutritional content and myth busting section could be too generic to stick or have content that isn't relevant to our end users at all.

The question of how to position nutritional information in mHealth application for MCH causes one to consider many aspects related to the spread and use of research-back nutritional information. Martin [10] conducted a systematic scoping review of 87 articles from 63 studies examining how social and behavioural interventions have involved family members in maternal, infant, and young child nutrition in low and middle income countries. The review aligns with findings by Till [17] that the influence of fathers, grandmothers, and other family members on maternal and child nutrition has been well documented, yet the majority of interventions continue to exclude them [10]. The review found that few interventions addressed gender norms, decision making, or family dynamics [10]. This is a critical gap for our project to address, as food taboos have been

seen to be primarily spread by family members and communities. [4] [15].

The three papers converge on the belief the information provided by mHealth interventions is unlikely to translate into changed behaviour unless the family members who influence her food choices are included. These findings reinforce the need for our project's multi-stakeholder approach, where nutritional content is not only delivered to mothers but is also accessible to family members.

3 Critical Comparison of Prior Work

The scoping reviews by Till [17] and Adusei-Mensah [1] both establish that mHealth interventions for maternal health in LMICs have been designed without meaningful community involvement, but they emphased different elements of the issues within the mHealth for MCH space. Till focused on the format of content delivery and advocated for custom mobile applications over SMS-based systems, while Adusei-Mensah placed greater emphasis on the lack of cultural and infrastructural consideration in current mHealth solutions. Our project will learn from and draw on the key issues highlighted by both. We will build a custom application that enables us to deliver culturally adapted multimedia content in an offline-first format, addressing both the infrastructural and contextual gaps that the reviews identify.

MomConnect's scale, enrolling over 494,000 women in its first year and reaching 60% of pregnant women at first antenatal appointments by 2017 [3], demonstrates strong demand for mobile maternal health content. Yet its reliance on USSD and WhatsApp has limited its functionality and increased costs [3] [14], reinforcing Till et al.'s argument that messaging-based systems cannot deliver the narrative culturally tailored content the literature calls for. Our project is perfectly positioned to address this gap.

Muthelo's research [12] is the most directly relevant study to our context, and it sits at the intersection of many themes raised across the literature. The barriers participants identified, including data costs, storage limitations, and poor connectivity, are consistent with the infrastructural challenges flagged by both scoping reviews. Their preference for video and audio content in a single integrated app aligns with Till's advocacy for custom applications and with Farao's argument for digital storytelling [7]. Muthelo's study captured what communities want, but did not test whether delivering their desires improves outcomes. Our project will hopefully serve as a starting point for answering this question.

The co-design literature differs in depth of community involvement. Coleman [5] conducted co-design workshops and confirmed that communities are open to digital interventions, so long as they are developed in their home language and fit their cultural context. The finding in this study that automated translation is inadequate for South African languages is a direct challenge for our project. Atukunda's work in rural Uganda [2] followed a more complete design cycle from interviews through to user acceptance testing, and the five elements for a successful application that participants identified provide a practical benchmark for our design. The overlap between Atukunda's findings and Muthelo's Limpopo workshops give strong guidelines for our project. We must ensure that it contains trustworthy information, accessible interfaces, and engaging content. Molefe and Dlamini [11] add quantitative backing to the findings of the reviews, showing that user-centered redesign improved adoption from 41.2% to 72.4% and nearly doubled 90-day retention in a South African mHealth context. The limitation is that their their study was not conducted in a rural or maternal health setting specifically, opening an opportunity for us to apply their methods in our project.

The design frameworks reviewed show promise in providing guidance for our development process, but neither provides a complete methodology that we could follow end to end. Farao's combined and adapted ISR and design thinking framework [8] proved effective for adapting a CHW-facing tool, but offered limited guidance on content development and does not emphasize co-design as strongly as the literature suggests is necessary. Nkwo's Persuasive Systems Design approach [13] provides compelling evidence to include features that incentivize healthy behaviours, and to be effective our project will need to utilize both.

Dsane-Nsor [6] and Felix [9] both show that mothers and parents actively evaluate health information based on its source, cultural relevance, and connection to their lived experiences. Both highlight the risk that even well-designed content will be may be dismissed by end users if it feels generic or culturally distant. Dsane-Nsor's finding that testimonials and narrative storytelling strengthen trust connects directly back to Farao's digital storytelling argument [7], and together they provide compelling evidence for our application to utilize community generated audio and video content.

The nutrition literature presents a clear gap for our rural Limpopo context. Chakona [4] and Ramulondi [15] researched and highlighted how food taboos inform dietary choices during pregnancy. Unfortuneatly, their findings are drawn from isiXhosa and Zulu communities, not from Sepedi-speaking populations in Limpopo. We will need to identify other sources to create the nutrition and myth-busting section of our application. Martin [10] confirms that family members significantly influence maternal nutrition choices, but most interventions target only mothers. Our project will acknowledge these findings and ensure the inclusion of other key family members.

4 Conclusion

The literature establishes a clear case for culturally adapted, community-focused mHealth interventions in low-resource maternal health contexts. Through research conducted in multiple African settings, the findings converge on the same core requirements: offline-first content delivery, multimedia content in local languages, inclusion of family members, and meaningful community involvement in design [17] [1] [12] [5]. Quantitative evidence confirms that designing for the end user's reality produces measurably better adoption and retention [11], and literature related to maternal health information sources makes it clear that content must feel culturally relevant and trusted to be acted upon [6] [9] [7].

The literature also surfaces tensions that our project will need to navigate. We will have to navigate between involving fathers and protecting the voices of women during our UAT testing [12] [16], balance cultural sensitivity and evidence-based health guidance in our multimedia content[4] [15], and define an achievable scope for an Honours project amongst all the needs the literature has identified. Ultimately, we hope that actioning on the insights of

researchers mentioned in this literature will allow us to deliver the first step of an mHealth solution that can effectively serve the women who need it.

References

[1] Frank Adusei-Mensah, Livhuwani Muthelo, Mxolisi Welcome Ngwenya, MH Mphasha, and Jussi Kauhanen. 2025. Digital health interventions for pregnant women and mothers with under 5-year-olds in low- and middle-income countries: a scoping review. *Global Health Journal* 9, 3 (2025), 113–123. doi:10.1016/j.glohj.2025.06.008

[2] Esther Cathyln Atukunda, Lynn T. Matthews, Angella Musiimenta, Godfrey Rwambuka Mugyenyi, Samuel Mugisha, Norma C. Ware, Celestino Obua, and Mark J. Siedner. 2021. mHealth-Based Health Promotion Intervention to Improve Use of Maternity Care Services Among Women in Rural Southwestern Uganda: Iterative Development Study. *JMIR Form Res* 5, 11 (2021), e29214. doi:10.2196/29214

[3] Peter Barron, Justin Peter, Amanda E. LeFevre, Jane Sebidi, Mia Bekker, Rachel Allen, Andrew N. Parsons, Peter Benjamin, and Yogan Pillay. 2018. Mobile health messaging service and helpdesk for South African mothers (MomConnect): history, successes and challenges. *BMJ Global Health* 3, Suppl 2 (2018), e000559. doi:10.1136/bmjgh-2017-000559

[4] Gamuchirai Chakona and Charlie Shackleton. 2019. Food Taboos and Cultural Beliefs Influence Food Choice and Dietary Preferences among Pregnant Women in the Eastern Cape, South Africa. *Nutrients* 11, 11 (2019), 2668. doi:10.3390/nu11112668

[5] Toshka Coleman, Sarina Till, Jaydon Farao, Londiwe Shandu, Nonkululeko Khuzwayo, Livhuwani Muthelo, Masenyani Mbombi, Mamare Bopape, Alastair van Heerden, Tebogo Mothiba, Shane Norris, Nervo Verdezoto, and Melissa Densmore. 2023. Reconsidering Priorities for Digital Maternal and Child Health: Community-centered Perspectives from South Africa. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2 (2023), 1–31. doi:10.1145/3610081

[6] Sarah Dsane-Nsor, Aakash Gautam, Carolina Fuentes, Gideon Kwesi Toffah, Nervo Verdezoto, Yaseen Joolay, and Melissa Densmore. 2025. Navigating Sociocultural Tensions in Prenatal Care: Opportunities for Digital Health from Diverse Stakeholders in Techiman, Ghana. *Computer Supported Cooperative Work (CSCW)* 34, 2 (2025), 741–784. doi:10.1007/s10606-025-09521-5

[7] Jaydon Farao. 2023. Digital Storytelling for the Support of Maternal and Child-wellbeing Through Community-based Co-design. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. ACM, New York, NY, USA, 1–5. doi:10.1145/3544549.3577050

[8] Jaydon Farao, Bessie Malila, Nailah Conrad, Tinashe Mutsvangwa, Molebogeng X. Rangaka, and Tania S. Douglas. 2020. A user-centred design framework for mHealth. *PLoS ONE* 15, 8 (2020), e0237910. doi:10.1371/journal.pone.0237910

[9] Beatriz Félix, Cristiana Braga, Xolani Ntinga, Sarina C. Till, Leina Meoli, Alastair Van Heerden, Ricardo Melo, Nervo Verdezoto, Melissa Densmore, and Francisco Nunes. 2024. Understanding How Parents Deal With the Health Advice They Receive: A Qualitative Study and Implications for the Design of Message-based Health Dissemination Systems for Child Health. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3643834.3661504

[10] Shannon L. Martin, Jennifer K. McCann, Emma Gascoigne, Dorothy Allotey, Daniel Fundira, and Kathryn L. Dickin. 2021. Engaging family members in maternal, infant and young child nutrition activities in low- and middle-income countries: A systematic scoping review. *Maternal & Child Nutrition* 17, S1 (2021), e13158. doi:10.1111/mcn.13158

[11] Thabo Molefe and Nolwazi Dlamini. 2025. Interactive Mobile Applications in Healthcare: A Comprehensive Analysis of User Experiences and Effectiveness. *Journal Mobile Technologies (JMS)* 3, 1 (2025), 31–38. doi:10.59431/jms.v3i1.542

[12] Livhuwani Muthelo, Masenyani Oupa Mbombi, Mamare Adelaide Bopape, Tebogo M. Mothiba, Melissa Densmore, Alastair van Heerden, Shane A. Norris, Nervo Verdezoto Dias, Paula Griffiths, and Nicola Mackintosh. 2023. Reflections on Digital Maternal and Child Health Support for Mothers and Community Health Workers in Rural Areas of Limpopo Province, South Africa. *Int. J. Environ. Res. Public Health* 20, 3 (2023), 1842. doi:10.3390/ijerph20031842

[13] Makuochi Samuel Nkwo, Rita Orji, Ifeyinwa Ajah, Joseph Igwe, Ignatius Ogbaga, and Chioma Chigozie-Okwum. 2021. Design Opportunities for Persuasive Mobile Apps to Support Maternal and Child Healthcare and Help-seeking Behaviors. In *Proceedings of the 3rd African Human-Computer Interaction Conference (AfriCHI '21)*. ACM, New York, NY, USA, 1–10. doi:10.1145/3448696.3448707

[14] Justin Peter, Peter Benjamin, Amanda E. LeFevre, Peter Barron, and Yogan Pillay. 2018. Taking digital health innovation to scale in South Africa: ten lessons from MomConnect. *BMJ Global Health* 3, Suppl 2 (2018), e000592. doi:10.1136/bmjgh-2017-000592

[15] Mmbulaheni Ramulondi, Helene de Wet, and Nontuthuko Rosemary Ntuli. 2021. Traditional food taboos and practices during pregnancy, postpartum recovery, and infant care of Zulu women in northern KwaZulu-Natal. *Journal of Ethnobiology and Ethnomedicine* 17 (2021), 15. doi:10.1186/s13002-021-00451-2

[16] Sarina Till, Jaydon Farao, Toshka Lauren Coleman, Londiwe Deborah Shandu, Nonkululeko Khuzwayo, Livhuwani Muthelo, Masenyani Oupa Mbombi, Mamare Motlhatlhedi, Molebogeng Motlhatlhedi, Gugulethu Mabena, Alastair Van Heerden, Tebogo Maria Mothiba, Shane Norris, Nervo Verdezoto, and Melissa Densmore. 2022. Community-based Co-design across Geographic Locations and Cultures: Methodological Lessons from Co-design Workshops in South Africa. In *Proceedings of the Participatory Design Conference 2022 - Volume 1 (PDC '22)*. ACM, New York, NY, USA, 120–132. doi:10.1145/3536169.3537786

[17] Sarina Till, Mirriam Mkhize, Jaydon Farao, Londiwe Deborah Shandu, Livhuwani Muthelo, Toshka Lauren Coleman, Masenyani Mbombi, Mamara Bopape, Sonja Klingberg, Alastair van Heerden, Tebogo Mothiba, Melissa Densmore, and Nervo Xavier Verdezoto Dias. 2023. Digital Health Technologies for Maternal and Child Health in Africa and Other Low- and Middle-Income Countries: Cross-disciplinary Scoping Review With Stakeholder Consultation. *J Med Internet Res* 25 (2023), e42161. doi:10.2196/42161